



New Heights Old Traditions

TAMILNADU COLLEGE OF ENGINEERING

The First Self-Financing Engineering College in Tamilnadu

Accredited with Grade 'A' by NAAC

(Approved by AICTE & Affiliated to Anna University, Chennai)

Estd : 1984

INTRUDER DETECTION IN SUREVILANCE

Under Guidance of :

maam's full name podanum

Assistant professor

Dept. of Computer science and Engineering

Tamilnadu college of Engineering

Coimbatore

Team members:

Abineeth TM (22CS04)

Basil V Mathew (22CS08)

Inbarasu M (22CS21)

Koushick G V (22CS29)

INTRODUCTION

In today's world, security is a top priority, and surveillance systems play a crucial role in maintaining it. However, traditional surveillance systems often rely on human monitoring, which can be prone to errors and inefficiencies. Our project aims to address this issue by developing an intelligent surveillance system that can detect individuals, recognize their faces, and identify their dress code in realtime, enabling swift action against potential security breaches. The proposed system has the potential to revolutionize the field of surveillance and security, providing a more effective and efficient solution for modern security challenges.

ABSTRACT

This project develops an intelligent surveillance system that detects individuals, recognizes their faces, and identifies their dress code in realtime, enabling swift action against potential security breaches. The system uses computer vision and deep learning techniques, including face detection, individual detection, and dress code recognition, to automate surveillance and reduce human error. By integrating these techniques, the system provides a robust and efficient solution for security personnel to monitor and respond to potential threats. The proposed system has the potential to enhance security measures in various settings, including public spaces, offices, and border control points, making it a valuable contribution to the field of surveillance and security.

Technology used

Language	: python
GUI	: tkinder
video processing	: cv2
face recognition	: face_recognition / numpy
object detection	: tensor flow
image processing	: pillow
data storage	: pickle / json /SQLite
file management	: os / pickle
alerts	: smtplib

Literature Survey

- **Face Detection:** MTCNN, RetinaFace, and Haar Cascade are state of the art techniques for face detection in surveillance systems.
- **Individual Detection:** YOLOv3/v4, Faster RCNN, and SSD are popular object detection algorithms used for individual detection.
- **Dress Code Recognition:** Deep-Fashion, Clothing-Net, and Transfer Learning approaches are used for dress code recognition.
- **Research Gap:** Limited studies combine face detection, person detection, and dress code recognition in real time CCTV streams, highlighting the need for an integrated system.
- **Key Takeaway:** Our project will leverage MTCNN/YOLOv3 for detection and finetuned CNNs for dress code, addressing realtime performance and robustness to CCTV constraints.

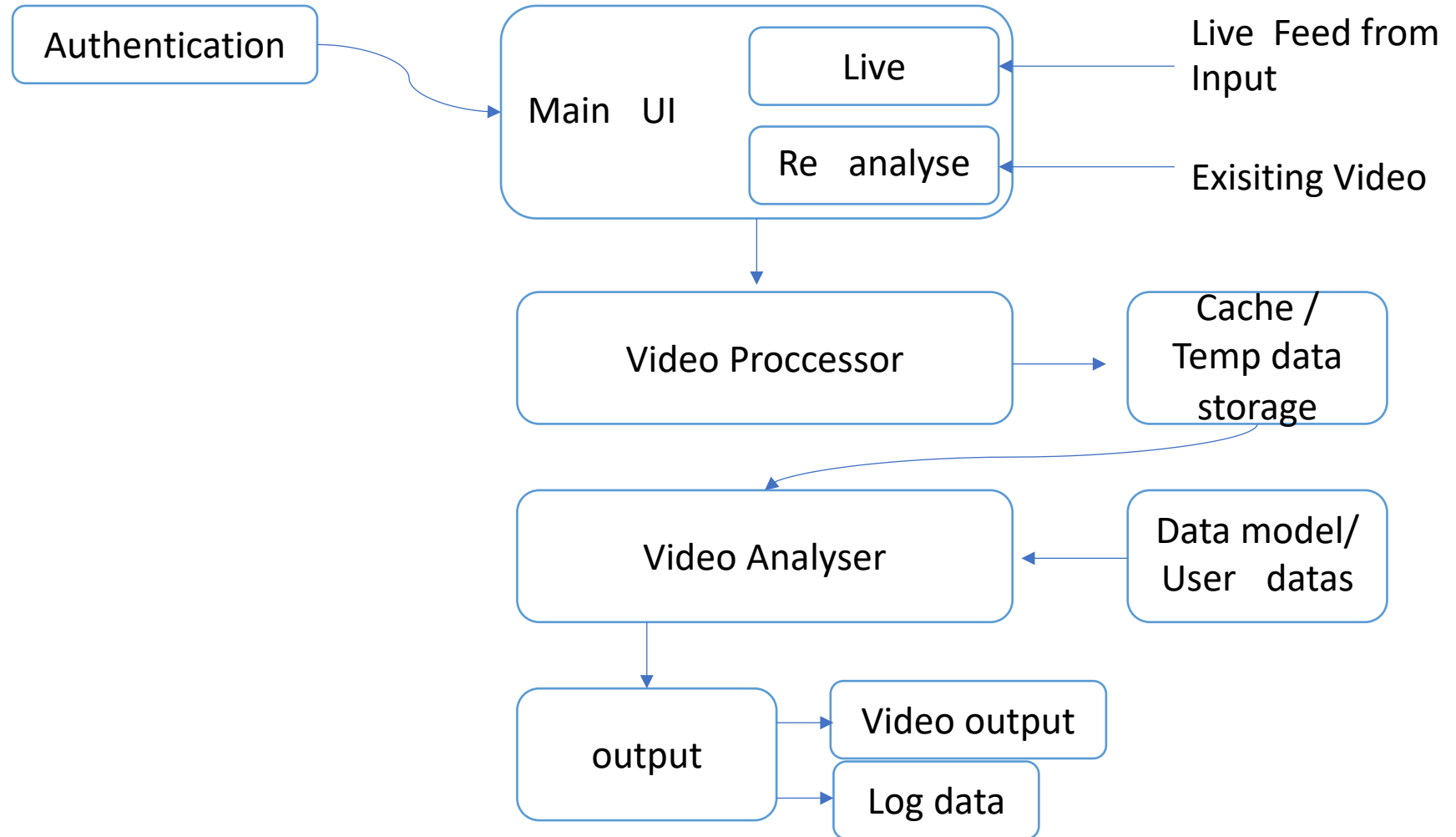
Existing System

- Security mainly relies on ID cards, RFID, or manual verification
- CCTV cameras are monitored manually by security personnel
- Biometric systems (fingerprint / RFID) are limited to entry points
- Face recognition systems (commercial) are costly and cloud-dependent
- No automatic intruder detection mechanism
- No verification of dress code compliance
- No visual check for ID card presence
- High dependence on human supervision
- Prone to human error, card misuse, and delayed response
- Limited automation, scalability, and customization

Proposed system

- An **automated security system** using computer vision and AI
- **Face recognition** to identify authorized employees
- **Intruder detection** for unknown or unauthorized individuals
- **Employee face enrollment** and secure data storage
- **Dress code detection** (e.g., shirt / T-shirt classification)
- **ID card presence detection** (visual check only)
- **Real-time monitoring** using camera feed
- **Automatic alerts and logging** of intruder events
- **Windows-based application** with user-friendly GUI
- **Offline, cost-effective, and scalable** solution

FLOW CHART



Usage of Project in Real-Time Environment

- Office & Corporate Buildings – Automated employee identification and intruder detection
- Restricted Areas – Labs, data centers, server rooms, and control rooms
- Educational Institutions – Secure access to staff-only zones
- Industrial Facilities – Monitoring entry compliance with dress code rules
- Hospitals & Research Centers – Controlled access to sensitive departments
- Government Offices – Enhanced security with minimal human supervision
- Small & Medium Enterprises (SMEs) – Cost-effective offline security solution
- Attendance & Monitoring Support – Employee presence verification
- Security Auditing – Intruder image capture and event logging

Core Research Papers

Doc ID / Link	Abstract	Additional Notes
2019_IEEE-ANTS https://dspace.nitrkl.ac.in	Video-based real-time intrusion detection system using deep learning methods (YOLO, SORT) on edge devices for automated surveillance and intruder alerts.	★ Most Relevant:Directly combines video processing + intrusion detection. Uses YOLO for object detection and SORT for multi-object tracking. Real-time edge device implementation.
IEEE-7852676 ieeexplore.ieee.org	Intrusion detection of specific area based on video using Gaussian Mixture Model (GMM) improvements for background subtraction and object detection/tracking.	★ Highly Relevant:Uses GMM for background modeling. Area-specific detection. Good for region-based intrusion alerts in surveillance zones.
IEEE-10134400 ieeexplore.ieee.org	Intrusion detection using camera and alert management. Image processing with Sum of Absolute Difference (SAD), camera-based motion detection and automated alerting.	Relevant:SAD method for motion detection. Good for simple frame-to-frame difference. Alert system integration.
IEEE-10370984 ieeexplore.ieee.org	Intelligent intruder detection system for surveillance rover using neural-network-based detection on mobile platform with autonomous movement capabilities.	Advanced:Mobile/drone-based detection. Neural networks for classification. Good for moving camera scenarios.

Conclusion

- Developed an automated security system using computer vision and AI
- Successfully integrates face recognition for employee identification
- Detects intruders in real time with image logging
- Verifies dress code compliance and ID card presence
- Reduces dependency on manual monitoring
- Works as an offline, cost-effective Windows application
- Suitable for small to medium secure facilities
- Can be extended with GPU acceleration, liveness detection, and multi-camera support
- The proposed system enhances security by combining identity verification with compliance monitoring in real-time.

References

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- https://en.wikipedia.org/wiki/Computer_vision
- https://en.wikipedia.org/wiki/Object_detection
- https://en.wikipedia.org/wiki/Convolutional_neural_network
- <https://en.wikipedia.org/wiki/Biometrics>
- <https://docs.opencv.org>
- https://github.com/ageitgey/face_recognition